

As a particle traveling through space, the ‘global’ origin at the point $(0, 0, 0)$ is not that useful. Oftentimes we don’t bother writing the origin/axes since it only serves to clutter the discussion. What *does* matter is your direction of motion ($\mathbf{T}$), the direction in which you turn ($\mathbf{N}$), and the tendency for your particle to leave the plane generated by $\mathbf{T}$ and $\mathbf{N}$. This plane is defined by the orthogonal vector

$$\mathbf{B} = \mathbf{T} \times \mathbf{N}.$$

This vector is known as the *binormal* vector.

These three vectors form a traveling-coordinate system—sometimes called a *frame*. This specific frame is called the **Frenet frame** or **TNB frame**.

Within this framework, we can re-envision the notion of acceleration. Frequently, for example, we want to know how much acceleration acts in the direction of motion. Recalling that, in Section 12.3 we showed that $\mathbf{T} = \frac{d\mathbf{r}}{ds}$, we start again with velocity:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \frac{d\mathbf{r}}{ds} \frac{ds}{dt} = \mathbf{T} \frac{ds}{dt}$$

Differentiating yields¹

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d}{dt} \left(\mathbf{T} \frac{ds}{dt} \right) =$$

¹In Section 12.3, we showed $\frac{ds}{dt} = \|\mathbf{v}\|$ and, in Section 12.4, we showed that $\mathbf{N} = \frac{1}{\kappa} \frac{d\mathbf{T}}{ds}$ so that $\frac{d\mathbf{T}}{ds} = \kappa \mathbf{N}$.

Definition. If the acceleration vector is written as

$$\mathbf{a} = a_T \mathbf{T} + a_N \mathbf{N},$$

then

$$a_T = \frac{d^2 s}{dt^2} = \frac{d}{dt} \|\mathbf{v}\| \quad \text{and} \quad a_N = \kappa \left(\frac{ds}{dt} \right)^2 = \kappa \|\mathbf{v}\|^2$$

are the **tangential** and **normal** scalar components of acceleration.

Visually, the parallelogram law allows us to ‘see’ acceleration direct itself between $\mathbf{T}$ and $\mathbf{N}$:

There are *many* different ways to visualize and compute a_T and a_N . The above picture suggests that we can think of a_T as the magnitude of $\text{proj}_{\mathbf{T}}(\mathbf{a})$ and a_N as the magnitude of $\text{proj}_{\mathbf{N}}(\mathbf{a})$ (i.e., the scalar projections—see Section 11.3). Moreover, since $\mathbf{T}$ and $\mathbf{N}$ are orthogonal, alternate interior angles and complementary angles state that $\sin(\theta) = \cos(\phi)$, where θ is the angle between $\mathbf{v}$ and $\mathbf{a}$ and ϕ is the angle between $\mathbf{N}$ and $\mathbf{a}$. Specifically,

$$a_T = \frac{\mathbf{a} \cdot \mathbf{T}}{\|\mathbf{T}\|} = \mathbf{a} \cdot \mathbf{T} = \mathbf{a} \cdot \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{\mathbf{v} \cdot \mathbf{a}}{\|\mathbf{v}\|}$$

and

$$a_N = \frac{\mathbf{a} \cdot \mathbf{N}}{\|\mathbf{N}\|} = \mathbf{a} \cdot \mathbf{N} = \|\mathbf{a}\| \|\mathbf{N}\| \cos(\phi) = \|\mathbf{a}\| \sin(\theta) = \|\mathbf{a}\| \frac{\|\mathbf{v} \times \mathbf{a}\|}{\|\mathbf{v}\| \|\mathbf{a}\|} = \frac{\|\mathbf{v} \times \mathbf{a}\|}{\|\mathbf{v}\|}$$

Observe that $\mathbf{B}$ is nowhere to be found! No matter how much our particle twists and turns, its acceleration will *always* lie in the plane defined by $\mathbf{T}$ and $\mathbf{N}$. Additionally, we note that

- a_T measures the rate of change of the *length* of $\mathbf{v}$. This is the change in speed.
- a_N measures the rate of change of the *direction* of $\mathbf{v}$.

Notice that computing a_N directly is more taxing than computing a_T . This comes about since a_N requires either the curvature κ or a cross-product. Thus, if you already know the final acceleration vector $\mathbf{a}$, sometimes it is easier to find a_T first and then solve for a_N . To see this, start by observing that

$$\|\mathbf{a}\|^2 = \mathbf{a} \cdot \mathbf{a} = a_T^2 + a_N^2.$$

Then,

$$a_N = \sqrt{\|\mathbf{a}\|^2 - a_T^2}.$$

Example. Without finding $\mathbf{T}$ and $\mathbf{N}$, write the acceleration of the motion

$$\mathbf{r}(t) = \langle 6 \cos(t), 6 \sin(t), 1 \rangle,$$

in the form $\mathbf{a} = a_T \mathbf{T} + a_N \mathbf{N}$.

Example. Without finding $\mathbf{T}$ and $\mathbf{N}$, write the acceleration of the motion

$$\mathbf{r}(t) = \langle \cos(t) + t \sin(t), \sin(t) - t \cos(t) \rangle, \quad t > 0$$

in the form $\mathbf{a} = a_T \mathbf{T} + a_N \mathbf{N}$